DSA4213 Assignment 1: Does better next-token prediction give better representations?

Language models, embeddings and downstream topic classification on WikiText-2 and AG News

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1 Question and setup

Two things are commonly assumed about small language models (LMs): that lower perplexity means a better model, and that better LMs give better text representations. This report tests both on one corpus. The comparison question is: *does better next-token prediction translate into better frozen representations for topic classification, and how much do input-embedding choices matter?* Part I trains four LM families on WikiText-2 under one recipe; Part II keeps the LSTM and varies only its input-embedding table in a 3×2 (source $\times$ trainability) design; Part III feeds the frozen models' states to one linear classifier on AG News. Everything not under study is held fixed: vocabulary, token stream, embedding width (300), output layer (untied), optimiser, schedule, early stopping, evaluation targets, prompts, decoding and classifier.

2 Data and common protocol

Corpus and vocabulary. WikiText-2 [1] (pre-tokenised, with its own `<unk>`) is lowercased once, one `<eos>` is appended per line, and the vocabulary is built from the train split only: 28,913 types. Lowercasing is a deliberate fairness decision: GloVe-6B is uncased, so a cased vocabulary would let *The*/*the* share one vector only in the GloVe setting of Part II. It shrinks the vocabulary from 33,278 types, so the perplexities below are not comparable with published WikiText-2 numbers. Validation and test contain no out-of-vocabulary tokens beyond the dataset's own `<unk>` (6.2% of test tokens), and `<unk>` is an ordinary vocabulary item for every model, including the n-gram.

Streams and perplexity. Each split is one continuous token stream (train 2,075,678 tokens, validation 216,348); no padding exists, so nothing has to be excluded from any loss. Every model is scored on the same 244,102 test targets, and perplexity is exp of the mean per-target cross-entropy. Recurrent models are scored with batch 1 and the hidden state carried over the entire stream (full history); the Transformer with a sliding window of 128 tokens moved by 32, scoring the last 32 targets of each window (each target sees $\geq$96 tokens of context). A second protocol, non-overlapping 128-token blocks with the state reset (identical for every model), isolates the effect of context length. The same protocol is used for validation during training, so the curves and the final numbers are directly comparable.

Models and recipe. All neural models share a 300-d input embedding initialised $\mathcal{U}(-0.1, 0.1)$ and a $300 \times V$ output layer, so they differ only in the sequence core: a 2-layer tanh RNN (0.36 M core parameters), a 2-layer LSTM (1.44 M), a 4-layer pre-LN Transformer with 6 heads and FFN width 1200 (4.37 M), and, as a parameter-count control, the same Transformer with 1 layer (1.09 M, close to the LSTM). The n-gram baseline is interpolated Kneser–Ney [2, 3] with one discount per order, orders 2–5 fitted and the order chosen on validation. The neural recipe is identical for every run: batch 32×128-token BPTT chunks with the state detached between chunks, dropout 0.3, Adam (10^{-3}, 200-step warm-up), learning rate halved after one epoch without validation improvement, early stopping after three, at most 12 epochs, global gradient-norm clipping at 1.0 with the pre-clip norm logged at every step. A unit test perturbs future tokens and checks that no model's logits at earlier positions change (causality), and that input and output tables are not tied.

Cost measurement. All runs use one Apple M4 (10 cores, 16 GB) through PyTorch's MPS backend, one process at a time. Timers synchronise the device before and after each measurement; epoch 1 (shader compilation) is reported separately from later epochs. Model size is the parameter count and the saved fp32 checkpoint. The common inference workload is (A) scoring the full test stream with the protocol above and (B) generating 50 tokens for each of 5 prompts at batch 1 with top-k sampling (the Transformer without a KV cache). MPS is not bit-deterministic; the seed-1 repeats in Part II give the run-to-run spread, about one perplexity point, and `-device cpu` reproduces exactly.

Model	Test PPL	Test PPL, ctx≤128	Val PPL, ep. 3	Val PPL, 8 min	Val PPL, ep. 12	Test PPL, 24 ep
KN 5-gram	214.9	–	–	–	–	
RNN	173.0	177.2	219.6	210.0	187.3	
LSTM	134.2	150.4	237.3	214.8	145.4	
Transformer (4L)	140.9	146.4	227.9	227.9	150.1	
Transformer (1L, control)	171.4	174.7	256.9	234.0	183.1	

Table 1: Part I prediction quality on the common 244,102 test targets. "ctx≤128" resets the state every 128 tokens for every model; the epoch-3 and 8-minute columns compare the models at a matched number of updates and at a matched wall-clock budget; the last column is a separate 24-epoch run of the same configuration (convergence check).

Model	Params (M)	Core (M)	Size (MB)	Train to best (min)	Score (tok/s)	Generate (ms/tok)
KN 5-gram	5.8 entries	–	156	0.1	318,385	19.7
RNN	17.7	0.36	71	23.2	6,779	1.6
LSTM	18.8	1.44	75	22.1	26,333	1.7
Transformer (4L)	21.8	4.37	87	29.8	12,694	3.4
Transformer (1L, control)	18.5	1.12	74	22.0	18,758	2.3

Table 2: Part I cost on one Apple M4 (MPS). "Core" excludes the identical embedding and output layers. Score: test-stream scoring throughput under the reported protocol; generate: batch-1 sampling.

Generation. Five prompts are the first 10 tokens of seeded-random *validation* lines (never test), and every model continues each prompt for 50 tokens with greedy decoding and with top-k=40, T=1 sampling from one shared CPU sampler, so decoding is matched exactly across the n-gram and neural models. Continuations are summarised by distinct-n ratios and the fraction of 4-grams that repeat an earlier 4-gram of the same continuation.

3 Part I: four language models on one corpus

Observations. *Prediction quality* (Table 1). With full history the order is LSTM (134.2) < Transformer (140.9) < RNN (173.0) ≈ 1-layer Transformer (171.4) < KN 5-gram (214.9). The order of the two best models *reverses* when every model's context is reset every 128 tokens: the Transformer loses only 5.6 points, the LSTM loses 16.2 and ends up behind it. *Learning behaviour* (Fig. 1). All four neural models were still improving at the 12-epoch cap: no learning-rate halving and no early stop fired, and the train/validation gap grows (LSTM: 113.1 vs. 145.4 at epoch 12) without validation loss turning up, i.e. no overfitting at this budget. Per epoch the LSTM and 4-layer Transformer track each other closely; per wall-clock minute the LSTM leads because its epoch takes 110 s against 149 s. After three epochs the tanh RNN is the *best* model (219.6 vs. 237.3 and 227.9); it learns fastest and saturates first. *Where the differences sit* (Fig. 2). On the 3% of test targets seen at most 10 times in training the n-gram is the best model (12.25 nats vs. 13.16 for the LSTM); every neural model wins on words seen ≥11 times and the margin widens with frequency (>10k bucket: LSTM 2.26 vs. n-gram 2.65). The Transformer is the best model on `<eos>` (1.09 vs. 1.30) and the worst on `<unk>` (3.51 vs. 3.19). *Cost* (Table 2). The 4-layer Transformer needs 29.8 min of training against 22.1 for the LSTM and reaches a worse full-history perplexity with 3× the core parameters. At inference the LSTM scores the test stream at 26,333 tokens/s against 12,694 for the sliding-window Transformer, and samples at 1.7 vs. 3.4 ms per token (no KV cache). The tanh RNN trains as fast as the LSTM but scores at only 6,779 tokens/s at batch 1. The n-gram fits in 4 s and scores at 318,385 tokens/s, but its 156 MB of tables exceed any checkpoint and sampling its full distribution costs 19.7 ms per token. *Generated text* (Table 3, Appendix A). Under greedy decoding the RNN and the 1-layer Transformer fall into loops (74% and 57% of 4-grams repeat an earlier 4-gram; n-gram 58%), while the LSTM and 4-layer Transformer repeat far less (25%, 31%). Top-k sampling removes repetition for every model (≤2.6%), so sampled continuations differ in local fluency and consistency instead: the LSTM and Transformer hold a topic across the 50 tokens (a "route description" section after the highway prompt, the Ming court after the 1644 prompt), the RNN drifts within a sentence and the n-gram within a few tokens.

Explanations, separated from the observations. *Context length, not architecture, explains LSTM > Transformer.* The reversal under the context-reset protocol is direct evidence: at ≤128 tokens the Transformer is the better predictor, and the LSTM's overall lead comes from information carried beyond 128 tokens (WikiText articles are thousands of tokens long). A Transformer with a longer window would plausibly close the gap; this was not tested. *Parameter count does not explain the ranking.* The 1-layer Transformer has about the LSTM's core size (1.12M vs. 1.44M) and is 37.2 points worse, while the 4-layer model has 3× the LSTM's core and is still worse with full history; depth matters for the Transformer, but more parameters did not buy a better model at this budget. *Training budget conditions every conclusion.* Nothing converged in 12 epochs and the ranking at epoch 3 (RNN first) differs from that at epoch 12, so the ordering could still change; the last column of

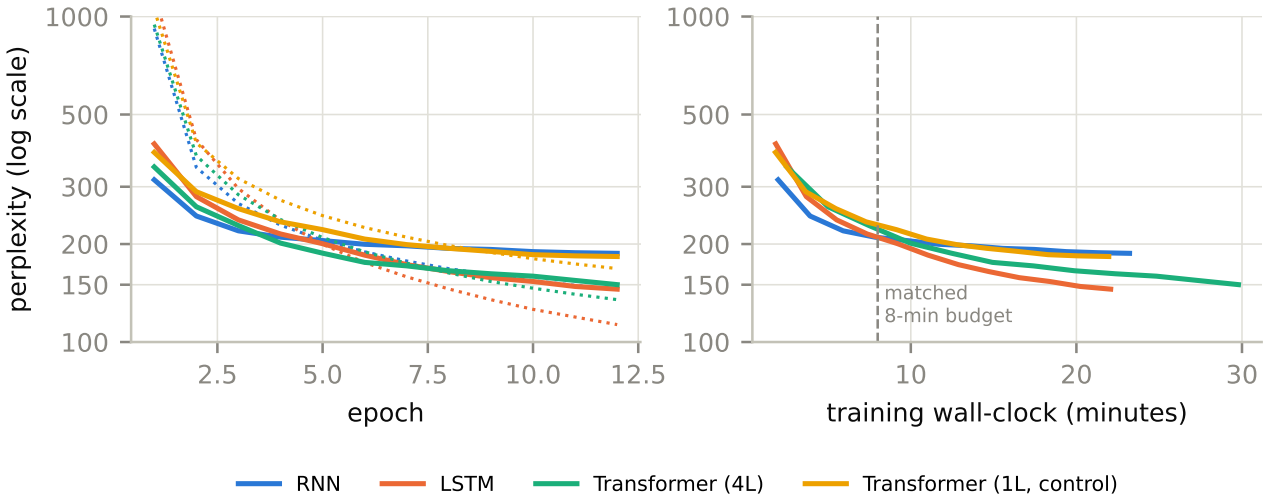


Figure 1: Validation perplexity (solid) and train perplexity (dotted) per epoch (left) and validation perplexity against training wall-clock (right); dashed: the separate 24-epoch runs.

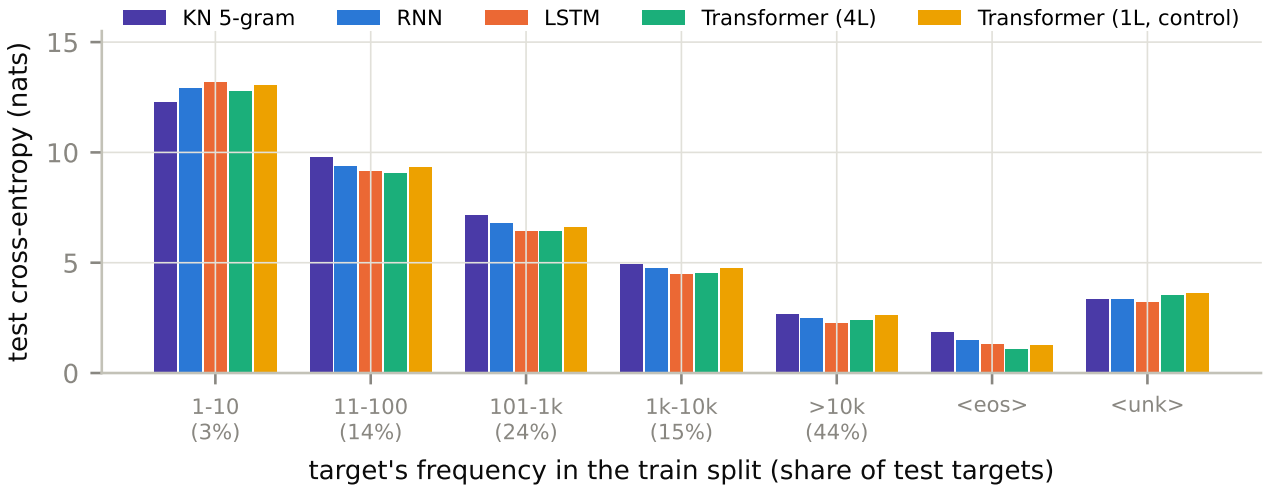


Figure 2: Test cross-entropy split by how often the target word occurs in the train split.

Table 1 reports a 24-epoch run of the LSTM and Transformer as a check (– vs. – test PPL). *The n-gram's edge on rare words* is expected from Kneser–Ney's continuation counts, which model how many contexts a rare type appears in, whereas an embedding row updated at most ten times has barely moved from its initialisation (a hypothesis; Part II tests it with pre-trained rows). *The RNN's slow scoring* is an implementation effect (no fused MPS kernel for batch-1 tanh recurrence), not an architectural one, and the greedy loops of the RNN and 1-layer Transformer are consistent with their weaker modelling of the recent context rather than with decoding, since sampling removes them.

4 Part II: input-embedding ablation on the LSTM

Design. The LSTM of Part I is retrained with its 300-d input table replaced by (b) skip-gram vectors trained by us on the WikiText-2 train stream only (gensim, negative sampling; 1.9 min on the CPU, reported separately from LM training) and (c) public GloVe-6B vectors [4], both kept entirely frozen; (a) is the Part I model itself. Because the required three cells confound *trainability* with *information content*, three cheap extra cells complete a 3×2 factorial: the same two pre-trained tables left trainable, and a frozen random table (no information at all). Skip-gram covers every type (`min_count`=1). GloVe covers 96.2% of the types and 94.74% of train tokens; the 1,104 missing types include the most frequent tokens of all (`<unk>`, `<eos>`, `@-@`, `@.@`, `@,@`, `'t`), and they are initialised $\mathcal{N}(0, \sigma_d)$ with σ_d the per-dimension standard deviation of the found rows (a frozen zero row for `<eos>` would cripple the model). The tables differ in scale: mean row norm 6.47 (GloVe), 2.37 (skip-gram) and about 1.0 for the random initialisation; nothing is rescaled, which is a disclosed confound. Freezing is verified by hashing the table before and after training. The required cells are run with seeds 0 and 1.

Model	Continuation (top-k=40, T=1, 50 tokens; first 24 shown)	distinct-2	rep. 4-gram
Prompt: `the americans knew of the approach of the japanese forces`			
KN 5-gram	`staged a sit in his home . <eos> with the rise of cobra commander in the sequel . <eos> he was appointed director @-@ ...`	0.98	0.00
RNN	`against the kpa 's king , and a peace @-@ force " it into a high position . <unk> had the right end of ...`	0.96	0.00
LSTM	`during the conflict . <eos> the first time against the american union were created by the royal army , both the republic of their ...`	0.96	0.00
Transformer (4L)	`and some of the song was " impossible " . <unk> said that " this were a " dramatic <unk> " and " ...`	0.90	0.00
Transformer (1L, control)	`against the 766th forces . <eos> the 766th headquarters was originally taken over them by the 766th regiment , 2nd division was appointed an ...`	0.94	0.00

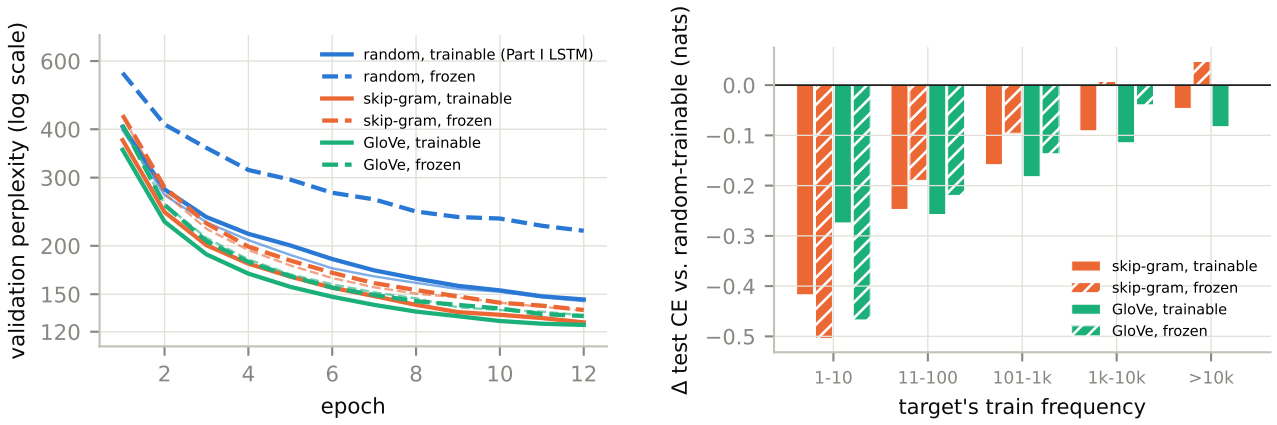
Table 3: Matched top-k continuations of one validation prompt (all prompts and greedy decoding in Appendix A). distinct-2: unique bigrams / bigrams; rep. 4-gram: fraction of 4-grams repeating an earlier one.

Embedding source	Trainable	Cov. (%)	Emb. cost	Test PPL	Val ep. 3	Val ep. 6	$\|g\|$ mean	Clipped	Row norm
random	yes	100.0	–	134.0 ± 0.3	237.3	184.9	0.27	0.3%	1.46
random	no	100.0	–	198.0	358.1	274.2	0.32	0.4%	1.00
skip-gram	yes	100.0	1.9 min	119.0	200.8	155.9	0.27	0.3%	2.71
skip-gram	no	100.0	1.9 min	127.9 ± 0.5	228.6	170.4	0.29	0.3%	2.37
GloVe	yes	96.2	public	116.5	190.1	147.7	0.27	0.3%	6.57
GloVe	no	96.2	public	123.8 ± 0.4	205.7	156.0	0.28	0.3%	6.47

Table 4: Part II: the LSTM of Part I with six input-embedding settings (rows 1, 4 and 6 are the required comparison; test PPL ± half the range over seeds 0 and 1 where two seeds were run). Cov.: vocabulary types with a pre-trained vector; "val ep. 3/6": validation PPL after 3 and 6 epochs (speed of convergence); $\|g\|$: mean pre-clip gradient norm; clipped: share of steps clipped; row norm: mean ℓ_2 norm of the table after training.

Observations. *Prediction quality* (Table 4). Both frozen pre-trained tables beat the jointly learned random table: skip-gram 127.9 and GloVe 123.8 against 134.0 test PPL, while the frozen random table collapses to 198.0. Letting the pre-trained tables train improves them further, to 119.0 and 116.5, the two best language models of this report. The two seeds of the random-trainable cell differ by 2×0.3 points, which is the noise floor; every contrast discussed below is several times larger except the skip-gram/GloVe difference among trainable tables (2.6 points). *Convergence* (Fig. 3a, columns "val ep. 3/6"). The ordering is already visible after three epochs (GloVe-trainable 190.1 vs. random 237.3 vs. random-frozen 358.1), and the frozen skip-gram and GloVe cells pass the random cell's final 12-epoch perplexity at epochs 10 and 8; the random-frozen cell never does. *Stability.* Mean pre-clip gradient norms (0.27–0.32), the share of clipped steps ($\leq$0.4%) and the within-epoch spread of the step loss are indistinguishable across cells; the only stability differences are isolated spikes (a maximum norm of 13.1 in one seed of the random cell and 5.8 for the frozen random table, against $\leq$2.3 elsewhere). *Generalisation.* Frozen tables fit the training stream less but generalise relatively better: the validation–train perplexity gap at epoch 12 is 13.7 (GloVe frozen) and 14.7 (skip-gram frozen) against 32.2 for the random table and 26.4/29.8 once the pre-trained tables train, consistent with 8.7 M fewer trainable parameters. *Where the gain sits* (Fig. 3b). The benefit of a frozen pre-trained table is concentrated on rare words: on targets seen at most 10 times in training, frozen GloVe and skip-gram cut the cross-entropy by 0.47 and 0.50 nats relative to the random table, the gain shrinks with frequency and vanishes on words seen more than 10k times (0.00 and -0.05 nats). Making the tables trainable shifts the pattern: trainable GloVe gains less on the rarest words (0.27) but is the only setting that also improves the most frequent ones (+0.08), and the frozen random table loses on every bucket. *Generated text* (Appendix A). Sampled continuations from the pre-trained cells are locally fluent WikiText (section headings, unit conversions such as "2 – 1 km (8 @.@ 7 mi)"), whereas the frozen random table degenerates into `<unk> @-@ <unk>` chains (top-k distinct-2 0.84 vs. 0.95 for the Part I LSTM). Under greedy decoding repetition is *not* monotone in perplexity: the best cells loop more often (46% of 4-grams for trainable GloVe, 41% for frozen skip-gram) than the Part I LSTM (25%), an observation these experiments do not explain.

Trainability versus extra data. The factorial separates the two factors the brief asks about. *Trainability* matters enormously when the table carries no information (frozen vs. trainable random: 64.1 points) and only moderately once it does (skip-gram: 8.8; GloVe: 7.3). *Information in the table* is worth more than trainability: replacing the trainable random table by a frozen skip-gram table trained on the very same 2 M tokens gains 6.1 points, for 1.9 min of CPU time (under a tenth of one LM run). *Extra data* (GloVe's 6 B tokens against our 2 M) buys a further 4.1 points frozen and 2.6 trainable, i.e. less than having any sensible pre-trained table at all. Two caveats pull GloVe's figures in opposite directions: its most frequent missing types (`<eos>`, `<unk>`, `@-@`) sit frozen at random vectors, a handicap, while GloVe-6B was trained on Wikipedia 2014, which almost certainly contains the WikiText-2 source articles; and the row-norm confound (6× longer rows) could explain part of its edge. Our reading, stated as a hypothesis: with 2 M tokens and 12 epochs the 8.7 M-parameter input table is the slowest-learning part of the model (rare rows receive few updates, cf. the n-gram's rare-word advantage in Part I), so any procedure that gives every row a usable vector before LM training starts is worth more than the



(a) Validation curves (thin: seed 1).

(b) Change in test CE vs. random-trainable by target frequency (hatched: frozen); the frozen random table, worse by 0.03–0.64 nats on every bucket, is omitted for scale.

Figure 3: Part II learning behaviour and where each embedding setting helps or hurts.

Representation	Dim	Source LM test PPL	Acc.	F1 (n=20,000)	F1 (n=2,000)	F1 (n=500)
LSTM, last state	300	134.2	62.7	62.3	59.4	54.5
LSTM, mean	300	134.2	68.1	67.9	64.2	60.6
Transformer, last state	300	140.9	73.6	73.6	69.9	66.6
Transformer, mean	300	140.9	77.7	77.6	73.9	69.5
RNN, mean	300	173.0	73.6	73.5	69.6	67.2
Transformer 1L, mean	300	171.4	79.0	78.9	74.8	70.9
LSTM + frozen skip-gram, mean	300	128.4	79.4	79.3	77.1	75.2
LSTM + frozen GloVe, mean	300	123.4	82.4	82.3	79.6	77.8
embedding only: LSTM-learned table	300	–	75.2	75.1	70.0	63.3
embedding only: skip-gram	300	–	83.4	83.4	81.1	80.1
embedding only: GloVe	300	–	87.9	87.9	87.2	85.6
TF-IDF uni+bigrams (reference)	98,555	–	89.9	89.9	84.6	78.3

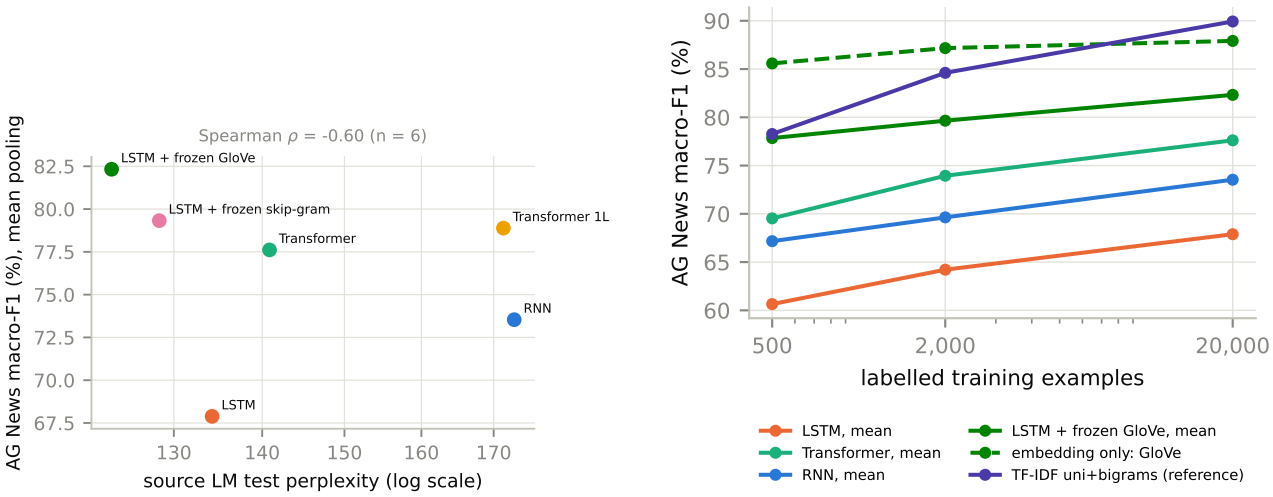
Table 5: AG News test accuracy and macro-F1 (%) of one multinomial logistic regression on each frozen representation, for 20,000, 2,000 and 500 labelled examples (C chosen on 2,000 validation documents).

freedom to adjust the rows during it.

5 Part III: representations for topic classification

Setup. Each AG News document (title + description, mapped to the WikiText vocabulary at 92.5% token coverage, `<eos>`-prefixed, truncated at 96 tokens) becomes one 300-d vector from a frozen model: the mean of the top-layer states over non-pad positions, or the state at the last real token; causal models never let padding reach a real position (unit-tested). One multinomial logistic regression on standardised features is fitted per representation, with C chosen on 2,000 held-out training documents and the 7,600-document test split used once. Embedding-only baselines (the mean of the raw table rows) and a TF-IDF reference over the same vocabulary ids isolate what the sequence model adds.

Observations. *Pooling.* Mean pooling beats the last state for both the LSTM (67.9 vs. 62.3 macro-F1) and the Transformer (77.6 vs. 73.6): the final state of a next-token model encodes what comes next, not what the document was about. *Models.* With mean pooling the Part I models rank 1-layer Transformer (78.9) > 4-layer Transformer (77.6) > RNN (73.5) > LSTM (67.9): the best language model is the worst feature extractor. *Embedding variants.* The LSTM with a frozen GloVe table gives 82.3 and with frozen skip-gram 79.3, far above the random table, but the raw tables alone do better still (GloVe 87.9, skip-gram 83.4), and even the LSTM's own learned input table (75.1) beats the LSTM's hidden states. *Data size* (Fig. 4b). TF-IDF is the best representation with 20,000 labels (89.9) but falls to 78.3 with 500, where mean GloVe still gives 85.6: dense pre-trained features dominate the low-label regime and sparse lexical features win once labels are plentiful; the LM-based representations sit below the embedding-only baselines at every size. *Errors* (Table 6, for LSTM + frozen GloVe, mean). Business and Sci/Tech are confused most often (526 of 7,600 test documents), then World and Business (285); the most confident errors are label noise, sports and stock-market stories filed under *World*. Vocabulary coverage does not explain errors: the `<unk>` rate is 7.5% in correctly and 7.4% in wrongly classified documents.



(a) Source-LM perplexity vs. downstream macro-F1.

(b) Macro-F1 against labelled-set size.

Figure 4: Does better next-token prediction give better representations?

true \ pred	World	Sports	Business	Sci/Tech
World	**1570**	102	145	83
Sports	68	**1762**	30	40
Business	140	52	**1430**	278
Sci/Tech	102	51	248	**1499**

True	Pred.	Conf.	Text (truncated)
World	Sports	1.00	Maddux Wins No. 302, Baker Wins No. 1,000 Greg Maddux pitched the Chicago Cubs into the lead in
World	Sports	1.00	Collingwood anchors England (AFP) AFP - Paul Collingwood's unbeaten 80 took England to 299 for
World	Sports	1.00	Football: Azerbaijan 0-1 England Michael Owen heads England's winner in the World Cup qualifier
World	Sports	1.00	Football: Brazil legend's UK debut Brazil football great Socrates is set to make his debut for

Table 6: Confusion matrix (left) and the most confident errors (right) of the best LM-based representation (LSTM + frozen GloVe, mean).

Does the LM ranking agree with the downstream ranking? No. Over the 6 LM-based mean-pooled representations the Spearman correlation between test perplexity and macro-F1 is $\rho = -0.60$ (Fig. 4a): lower perplexity goes with *worse* classification. The disagreement has two regimes. Across architectures with the same input table, perplexity and F1 move in opposite directions (LSTM best LM, worst features). Across input tables with the same architecture, the three LSTM cells improve together in both metrics, but both improvements come from the table, and the LSTM's states keep only part of what the table provides (82.3 vs. 87.9 for the raw GloVe mean). Our explanation, a hypothesis consistent with all three parts: a next-token objective on Wikipedia prose rewards local syntax and the last few words, so the state of a well-trained recurrent LM is specialised away from topic; the Transformer's residual stream carries the input embeddings to the top layer and the shallower Transformer carries them more faithfully (1 layer $>$ 4 layers), whereas LSTM gates overwrite them. Domain shift (lowercased Wikipedia to news, 92.5% coverage) hurts every LM-based representation alike and cannot produce the reversal; fine-tuning might, but was outside the frozen-representation question.

6 Conclusions, limitations and use of AI tools

On this corpus better next-token prediction did *not* give better frozen representations, and the input embedding table mattered more than the sequence model for both tasks. The LSTM is the best full-history predictor only because it carries state beyond 128 tokens; at equal context the 4-layer Transformer is better at a third more training time and half the scoring throughput. A frozen table trained on the same 2 M tokens in 1.9 min beats jointly learned embeddings, GloVe adds a modest further gain, trainability a little on top, and the gains sit on rare words. Downstream, mean-pooled raw GloVe (87.9) beats every LM-based feature and TF-IDF wins with enough labels (89.9). *Limitations.* One corpus, small models and a 12-epoch budget at which nothing had converged (the 24-epoch check in Table 1 bounds this); MPS training is not bit-reproducible (seed-to-seed spread about one perplexity point); GloVe's training data overlaps WikiText's source articles and its row norms differ from the other tables; Part III uses frozen models only, and AG News carries visible label noise. *Use of AI tools.* Claude Code (Anthropic, Claude Fable 5.1) [5] drafted the experiment design, the code, the scripts that generate every table and figure from the result files, and edits of this text; all experiments were run by the author and the analysis is the author's own (see `AI_USAGE.md`).

References

[1] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017.

[2] Reinhard Kneser and Hermann Ney. Improved backing-off for m-gram language modeling. In *ICASSP*, 1995.

[3] Stanley F. Chen and Joshua Goodman. An empirical study of smoothing techniques for language modeling. Technical report, Harvard University, TR-10-98, 1998.

[4] Jeffrey Pennington, Richard Socher, and Christopher D. Manning. GloVe: Global vectors for word representation. In *Proceedings of EMNLP*, 2014.

[5] Anthropic. Claude code (model claude fable 5.1). `https://claude.com/claude-code`, 2026. Coding assistant used as described in the accompanying AI_USAGE.md.

A All generated continuations

Prompt 1 the americans knew of the approach of the japanese forces

KN 5-gram, greedy in the region . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> ,

KN 5-gram, topk staged a sit in his home . <eos> with the rise of cobra commander in the sequel . <eos> he was appointed director @-@ <unk> during the late <unk> . another such cruise ship , loaded with troops , an individual 's perception have suggested that the team . the

RNN, greedy . <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos>

RNN, topk against the kpa 's king , and a peace @-@ force " it into a high position . <unk> had the right end of the 766th (" <eos> <unk> de <unk>) from the huai (emperor <unk>) in the cathedral area with the cathedral . <eos> the

LSTM, greedy . <eos> = = = = = = = = = = = = = = <eos> the first time of the year , the first time , was the first time of the year , and the first time was the first time to be the first time

LSTM, topk during the conflict . <eos> the first time against the american union were created by the royal army , both the republic of their <unk> against the country , and one of the 1st century ce , although the general government was supported by his party . the first two

Transformer (4L), greedy . <eos> = <eos> the song was written by the song by the song , and the song was written

Transformer (4L), topk and some of the song was " impossible " . <unk> said that " this were a " dramatic <unk> " and " <unk> ' and " <unk> " for the country ; the feast of their <unk> " was a standard word that was a major <unk> of

Transformer (1L, control), greedy . <eos> = = = = = <eos> the 766th regiment was the 766th regiment was the 766th regiment was the 766th regiment , and the 766th regiment was sent to reinforce the 766th regiment . the 766th regiment was sent to the 766th regiment , commanded by the 766th

Transformer (1L, control), topk against the 766th forces . <eos> the 766th headquarters was originally taken over them by the 766th regiment , 2nd division was appointed an <unk> unit of the <unk> , who had no advantage of fighting . these were lost after the 766th unit did not return to the 5th

Prompt 2 the ruins lie among the tropical rainforests of northern guatemala

KN 5-gram, greedy , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> , and the <unk> of the <unk> ,

KN 5-gram, topk , and was replaced by the national library , which was a large @-@ scale <unk> paintings and drawings . a few months later a bomb . <unk> of <unk> , with the <unk> in an attempt to play in the fa cup , but the soviet archives of dumont

RNN, greedy , and <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk> <unk> , <unk>

RNN, topk , and the middle section . <unk> for <unk> and the other group of the <unk> are used as a source of the georgian spacing is in the same . <eos> = = = early sources = = <eos> = = critical reception = = <eos> while critics were given

LSTM, greedy . the first two years later , the first of the first time was the first time in the first season , and the first time of the year , the <unk> of the <unk> , was the first time to be the first time . <eos> = = =

LSTM, topk , and the capital of the north end , which is not as a second <unk> and the east and west west of the southern gulf of the area , and it was the last five years of the united states during the first period , until september 14 ,

Transformer (4L), greedy . the north of the south of the east of the east of the south of the south korean peninsula , the south korean river , and the south korean peninsula were constructed in the north korean peninsula . the 766th was also built in the north korean area .

Transformer (4L), topk , the east 's western coast . the first few species of the west of <unk> were named as the site of the northern hemisphere and the north america , and south of <unk> and a small group of a. hygrometricus ; it was recorded . most recent species had

Transformer (1L, control), greedy , and the island of the island . <eos> = = = = = <eos> the storm was named after the storm , the storm was named after the storm , the storm was named after the storm , the storm was named after the storm , the storm was

Transformer (1L, control), topk , the island 's cathedral . <eos> = = = = <eos> the east of <unk> (<unk>) = = = <eos> = = development = <eos> the new york times the first period = = <eos> the next year) = <eos> = = reception = = =

Prompt 3 in 1644 , hu took it upon himself to create

KN 5-gram, greedy a new york times , the song is a re @-@ interpretation of the hymn " joyful , joyful " , and " <unk> " . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk>

KN 5-gram, topk a map of the poem 's the country 's " little to no use of this mushroom as if not exactly jammed together . the new yorker , and later , already putting a bit more polished <unk> <unk> , was also the first time in his career . he

RNN, greedy a new <unk> , and the <unk> of the <unk> . <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = = = = = <eos> = =

RNN, topk a suitable game . " , the player is the largest and other members of the united states as " its course " , and the belgian company , said in its existence of " one " <eos> <unk> of the city , " and the second of a year

LSTM, greedy a " <unk> " . <eos> = = = = = = = = = = = = = = <eos> = = = = = = = = = = = = = = = <eos> the first time of the film was the first time of

LSTM, topk a key of the school to the house of <unk> that was also used in the area . he also suggested that his life was to seek to a lot of education between the country and their <unk> <unk> , a number of six men . the <unk> @-@ off

Transformer (4L), greedy a <unk> of the jin . the song was the song by the jin , and the jin was the song . the song was the song of the song , and the song was the song by the song . the song was the song by the song song

Transformer (4L), topk a letter of the emperor and the jin would be attacked . the court of modern rule was forced to defend a jin . <eos> the huai river , although the jurchens attempted to secure a treaty , the song had been <unk> to hold the jin , this would

Transformer (1L, control), greedy a new state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of the state of maryland

Transformer (1L, control), topk a budget of 5 @,@ 000 in 2011 . <eos> after the next year , he has been awarded a special candidate for the state of yugoslavia . <eos> there is no doubt that he may have the <unk> , as a <unk> of a year of <unk> in the

Prompt 4 as the highway approaches thompson , us 2 leaves the

KN 5-gram, greedy area . <eos> = = = = <eos> the first recorded in the inventory of <unk> <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk> , a <unk>

KN 5-gram, topk house and <unk> <unk> . the first four starts , and the <unk> <unk> of the hindu in a single . <eos> meanwhile , a 2 @-@ 3 . a total of 15 @,@ 000 @,@ 000 acres (2 @.@ 3 % in the area , and was an

RNN, greedy east of the <unk> <unk> . <eos> = = = <unk> = = = <eos> the kakapo is a <unk> of the <unk> <unk> (<unk>) , and <unk> (<unk>) , <unk> <unk> (<unk>) , <unk> <unk> (<unk>) , <unk> <unk> (<unk>

RNN, topk east state . <eos> = = = = tropical storm developed the season , a tropical wave the system peaked in the azores and a storm . the storm caused wind gusts to a high winds of 65 mph) . <eos> = = = = <eos> a large number

LSTM, greedy <unk> of the <unk> . <eos> = = = = <unk> = = = = <eos> the first part of the first season was the first time of the season , and the first time was the first time to be the first time . <eos> = = = =

LSTM, topk area in the first area of the final half . the first of the south atlantic ocean , an additional , <unk> bridge , is in the end of the northern atlantic 's west . the road ' north of the city is at the top the west and the

Transformer (4L), greedy east of the east coast of the united states . <eos> = = = route description = = <eos> the highway is a highway between the highway of the highway system . the highway is located in the north of the north of the north of the north of the

Transformer (4L), topk east) . <eos> = = = route description = = <eos> the route has been laid in a park , <unk> avenue and the state state route in the village of the east . the highway continues north and south . it is a few <unk> road in the

Transformer (1L, control), greedy highway . <eos> = = = = <eos> the highway is a highway system to the highway system . the highway was designated as the highway system as a highway system of the highway system . the highway was designated as the highway system . the highway was designated as

Transformer (1L, control), topk system to the first line and , with an early 20th century , and by a village , was completed . <eos> hakim is used in the united states and made from a second @-@ day @-@ old film . <eos> = = = = = <eos> = = =

Prompt 5 churchill , ward (1996) . from a native

KN 5-gram, greedy `american tribes , the <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> , the <unk> <unk> ,`

KN 5-gram, topk `plant , as they were on the next to the player . " <unk> " and that his personal health , bethesda , maryland , in its original form of the god apollo bunder in the uk still " . in a review for the <unk> ) in a large`

RNN, greedy `american <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk> , <unk>`

RNN, topk `america , which were not known as , <unk> , tintin , the <unk> of the dominican republic and ireland , frelimo , which was held on the site of the jin dynasty college of war ii ( <unk> ) . <eos> " this is <unk> by a new york`

LSTM, greedy `american academy , the <unk> of the <unk> , was the first time of the first time . the first time was the first time of the year , and the first time was the first time to be the first time . <eos> = = = = = =`

LSTM, topk `newspaper , he had not been built that the other troops in the state is " just so what they had happened of all the life and a ] " . he concluded what it was born during the war . <unk> at a further <unk> from the british post`

Transformer (4L), greedy `american newspaper , the film was published in the united states . <eos> = = = production = = <eos> the film was released in september 2011 . the film was released in september 2011 . the film was released in september 2011 . the film was released in`

Transformer (4L), topk `sentence , wheeler 's book was known as the " good and " the first year that year was not the finest of one of the best film of the united kingdom during 1996 . in 2000 of the united states , wheeler continued to <unk> his wife . he`

Transformer (1L, control), greedy `american academy ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> ) , <unk> ( <unk> )`

Transformer (1L, control), topk `korean , which were also described as " <unk> " ) , a new school " ( 1999 ) . <eos> tintin in april 2010 , 2005 , " <unk> " ) : written by james f. kennedy , and wrote that this period in <unk> that year . in`

Model	Decoding	distinct-1	distinct-2	distinct-3	repeated 4-grams
KN 5-gram	greedy	0.24	0.36	0.39	0.58
KN 5-gram	topk	0.71	0.98	1.00	0.00
RNN	greedy	0.13	0.20	0.23	0.74
RNN	topk	0.64	0.91	0.97	0.01
LSTM	greedy	0.30	0.50	0.65	0.25
LSTM	topk	0.66	0.95	1.00	0.00
Transformer (4L)	greedy	0.30	0.48	0.61	0.31
Transformer (4L)	topk	0.63	0.94	0.99	0.00
Transformer (1L, control)	greedy	0.20	0.31	0.38	0.57
Transformer (1L, control)	topk	0.63	0.87	0.93	0.03

Table 7: Repetition statistics averaged over the five prompts.

B Embedding nearest neighbours

C Full hyper-parameters and reproduction

Reproduction: `run_all.sh` in the repository re-creates every number (about 5 h on the M4); `README.md` documents data access, preprocessing, dependencies, commands and seeds.

Word	Skip-gram (own, WikiText-2 train)	GloVe-6B
`king`	`veera, leinster, ballala, throne, vassal`	`queen, prince, monarch, kingdom, throne`
`city`	`metropolitan, borough, town, conurbation, burnside`	`cities, town, downtown, mayor, area`
`war`	`outbreak, airbase, italo, civil, hawaiians`	`wars, conflict, battle, civil, occupation`
`album`	`daydream, yankovic, charmbracelet, rca, slayer`	`albums, song, ep, band, songs`
`1990`	`1989, pr, oceania, sap, pursuant`	`1991, 1992, 1987, 1989, 1986`
`north`	`south, west, east, southward, alaska`	`south, east, korea, northeast, west`
`played`	`playing, striker, staged, matches, shortstop`	`playing, plays, play, player, matches`
`@-@`	`93rd, 10m, amassing, defamation, dash`	`channels, regions, beneficial, distress, cable`

Table 8: Five nearest neighbours (cosine) of probe words in the two pre-trained tables.

Setting	Value
Embedding dim / Transformer d_{model}	300
RNN / LSTM	2 layers, hidden 300, tanh / LSTM cells
Transformer	4 layers (control: 1), 6 heads, FFN 1200, pre-LN, GELU, learned positions, context 128
Dropout	0.3 after embedding, between layers, before output
Batch × BPTT	32 × 128 tokens (continuous stream, state detached between chunks)
Optimiser	Adam, lr 0.001, β=(0.9, 0.98), 200-step linear warm-up
Schedule	halve lr after 1 epoch(s) without val. improvement; early stop after 3; max 12 epochs
Gradient clipping	global norm 1.0
Output layer	untied (frozen input tables would otherwise freeze the softmax)
Validation / test protocol	recurrent: full history, batch 1; Transformer: sliding window 128 / stride 32
Decoding	greedy and top-k=40, T=1.0, 50 new tokens, seed 1234+prompt index
n-gram	interpolated Kneser–Ney, orders 2–5 fitted, order chosen on validation, $D_n = n_1/(n_1 + 2n_2)$
Skip-gram	gensim, sg=1, window 5, 10 negatives, subsampling 10^{-4}, 10 epochs, min_count 1, 1 worker
GloVe	6B tokens, 300d, uncased; missing rows $\mathcal{N}(0, \sigma_d)$ from the found rows
Part III classifier	scikit-learn multinomial logistic regression (lbfgs), standardised features, $C \in \{0.01, 0.1, 1, 10\}$
Seeds	0 everywhere; Part II required settings also seed 1

Table 9: Every setting shared by the runs; per-run differences are the model/embedding blocks in `configs/`.